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Application of Artificial Intelligence in Traffic Control System of Non-autonomous Vehicles at Signalized Road Intersection.

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Abstract

The increase in both rural and urban road traffic flow in recent years has led to several disasters in the transportation sector which include traffic congestion, accidents, and high rate of pollution. Alternative traffic control measures are needed whenever there is failure of conventional traffic control or real time traffic issues at road intersection. This current study seeks to investigate the stability and efficiency of Artificial Intelligence (AI) techniques, the artificial neural network (ANN) for eliminating or reducing traffic volume in the case of non-autonomous vehicles in a mixed South African traffic flow conditions. Electronic traffic data of one hundred and twenty six (126) vehicles were observed from Mikros Traffic Monitoring (MTM) firm, a subsidiary of Syntell Group of Company, South Africa. The traffic data was obtained via the traffic technologies employed at MTM which are basically sensor embedded on road surfaces to monitor and control vehicles which passes the traffic counter daily. The dataset obtained from MTM was trained, tested and validated using artificial neural network model under signalized road intersection in heterogeneous condition by using the class description of the vehicles, and corresponding speed as input variables. After series of training, the results suggest that ANN model produced the best possible results for traffic congestion in a heterogeneous traffic condition.

Keywords: Non-autonomous Vehicles; Artificial Intelligence; Artificial Neural Network

1. Introduction

In South Africa, as the human population continue to increase at an alarming rate there is a need for efficient, effective and robust road design and smooth transportation system with a balanced traffic control network architecture. Moreover, there is a high demand for suitable traffic congestion solutions that will positively benefit the lives of people. In the 2011 Population Census, the number of inhabitants in South Africa increased by 7 million between 2001 and 2011[1].

The problem of traffic congestion in road transportation has become an important real time transportation problem in developed and developing countries. Assessing and eliminating of this problem requires the application of efficient intelligent transportation techniques which has been introduced and have been effective at least to some extent in the last few years[1].

Transportation plays an imperative role in the lives of human being, nowadays the rate at which the human population is increasing and also the massive migration of people from rural to urban is threatening road infrastructures in metropolitan cities[2]. According to a documented report, in 2010 the United States of America spent about $115 billion in tackling traffic congestion in its 439 urban areas[3].

This population growth requires cogent solutions to combat population explosion without compromising the quality and effectiveness of transportation services. Among these is the need for effective transportation frameworks and arrangements that will reduce the traffic challenges that urban communities in South Africa are experiencing on daily [4].

According to the United Nations, South Africa is experiencing

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an increase in urbanization that may leave 71.3% of South African populace living in cities by 2030, which is expected to rise to 80% by 2050. Although, there is a consistent demand for road transportation needs in cities such as Johannesburg, Durban, Cape Town, and Pretoria. If there is a dilapidated public transportation system, people will resort to their private vehicles, which has resulted in different road transportation challenges including that of congestion and traffic car accidents[5].

Furthermore, [5] proposed that an increase in traffic gridlock is caused by higher levels of urbanization, population growth, changes in population density are the factors that results in challenges for the road transportation system as against the needs for efficient mobility of people and goods and also to create a sustained transportation system in South African Cities. Not only do these factors delayed the infrastructural development of the cities, but they also result in car accidents, traffic congestion, and increase in travel times, fuel consumption and carbon emanations.

Nonetheless, traffic blockage is seen to be a noteworthy problem that obstructs supportable street transportation in urban areas of South Africa. Although, a few traffic control measures have been taken that includes the introduction of public transportation, for example, Gautrain and Bus Rapid Transit System (BRT), in Johannesburg and Pretoria, and MyCiTi transport framework in Cape Town, Some parts of the urban areas are still experiencing transportation challenges. The urban areas, which doesn't have any public transportation framework, may face severe traffic blockage, during the pinnacle hours of the day. Therefore, it is argued that there is a need to investigate alternative roads that could ease traffic clog in Johannesburg.

According to the United Nations, South Africa is experiencing an increase in urbanization that may leave 71.3% of South African populace living in urban communities by 2030, which is expected to reach nearly 80% by 2050 [5]. Although, there is a demand for travel needs in major cities in South Africa, cities such as Johannesburg, Durban, Cape Town, and Pretoria. In the wake insufficient public transportation system in several major cities, people resort to self-propelled private vehicles, which has resulted to different road transportation challenges including that of congestion and traffic car accidents[5].

Over the past few years, a lot of research has been carried out on artificial intelligence in transportation systems, mostly on how it can be used to predict traffic flow on highways and its application in reducing traffic congestion. But, not much has been done in the areas of traffic congestion at a signalized road intersections especially if the vehicles at this intersections are autonomous vehicles.

Autonomous Vehicles are the future of road transportation, according to documented evidence, autonomous vehicles will help reduce traffic congestion by at least 50%. It will also go a long way in removing or reducing the application of gap acceptance theory in Un-signalized road intersections, in addition to reducing major accidents on the roads because in hazardous situation autonomous vehicle can react quicker than a non-autonomous vehicle due to their computational Intelligence.

From existing literature reviews, the use of artificial neural network (ANN) in Predicting traffic flow is not novel. However, previous related research neglected the application of artificial intelligence in the area of road intersections especially in signalized road intersection. This conference paper tried as much as possible to torchlight this aspect of road intersection. This paper focuses majorly on:

- Signalized Road Intersection.
- Using Artificial Neural Network to curb traffic congestion at signalized road intersection.

2.0 Autonomous Vehicle

An autonomous vehicle is an important part of artificial intelligence in transportation. The operation of an autonomous vehicle is taken over completely or partly by the vehicle itself. Autonomous vehicles can also be called "self-driving vehicle". There are so many advantages of using autonomous vehicles on roads. According to transportation engineers, these are the advantages of using them[6]:

- Introduction of autonomous vehicles on highways, freeways and road intersections will reduce traffic congestion by at least 50%.
- Autonomous vehicles can quickly avoid accidents on the road than a human driver due to their computational intelligence
- When the level of congestion is low, the speed of autonomous vehicles will also reduce. Moreover, due to their speed, efficiency and speed control mechanisms, autonomous vehicles will improve its energy efficiency by 20%.
- Autonomous vehicles also provide a good traveling experience.

All the advantages stated above should be enough reasons why autonomous vehicles should be introduced on the highways.

Although all the advantages are as a result of researches and theoretical studies that have been carried out on autonomous vehicles. Just by looking at the difference between real-life situations and modeling of systems which are related to autonomous vehicles. It is quite hard to understand the relationship between traffic congestion and autonomous vehicles. A lot of factors will dictate the compatibility of autonomous vehicles on the roads. Therefore, understanding the effect that autonomous vehicles have on real-life scenario will go a long way in helping researchers understand why the application of artificial intelligence is important in transportation networks[6].

Although traffic rules and regulations for controlling traffic flow at intersections have been implemented in so many developed countries by their respective governments, the evaluation and analysis of various strategies at real-life locations is not easy due to different issues such as legal restrictions, user acceptance, availability of unlimited possibilities, non-limitations of the controllers and financial constraints [6].

According to [5], Autonomous vehicles create a lot of opportunities such as making use of the significant advantages of the sensor technologies and artificial intelligence that is used to reduce human driving mistakes on road networks. Back in the

This is a resupply of March 2023 as the template used in the publication of the original article contained errors. The content of the article has remained unaffected.

olden days, when the only mode of transportation in the city of New York was still horse-powered vehicles, a National Conference on City Planning was organized and called off later because city planners were not able to find ways on how to reduce traffic congestion and health problems caused by horse-powered vehicles.

However, less than two decades after this conference was called off, a new invention in transportation was discovered called Internal Combustion Engine (also known as Horseless Carriage) was introduced into the transportation sector. This new invention called the Internal Combustion Engine reduced the animal odors and feces and to some extent traffic congestion[5].

In the early 1900s, the national conference on city planning that was supposed to be held in Washington DC was called off because of the inability of city planners to resolve traffic and health-related issues associated with the mode of transportation at that current time. In 1909, big urban cities like New York were over-reliant on vehicles that are powered by horses, even though these horse-powered vehicles are causing a big burden on the environment. After two decades, this problem was resolved without any interference from city planners or city designers. Some city planners have a cogent understanding of how powerful internal combustion engine vehicles will improve the transformation of cities (especially cities in developed countries) and the movement of people from one location to another[7].

Nowadays, after more than one decade that vehicles have transformed our environment and the way we move from one place to another, innovative technology (autonomous vehicle) can reshape and free how our intuition, the way we plan our cities and the design of cities urbanization. Autonomous vehicles are starting to be hitting our roads; some developed countries even have a target of 2050 for all their vehicles to be autonomous. Transportation engineers and city planners have already predicted that autonomous vehicles will change urban mobility setting in the not too distant future[7].

In 2013, the US Department of Transportation's National Highway Traffic Safety Administration (NHTSA) defined autonomous vehicles in six major ways[8]:

- **Zero**: When human drivers have full control of the vehicle.
- **One**: Some certain responsibilities, for example, controlling the steering wheel of the vehicle and vehicle acceleration are controlled without any human interference.
- **Two**: Vehicle function that is dependent on the immediate environment of the vehicle, in some instances the driver should be ready to take control of the vehicle.
- **Three:** Vehicles are fully 'self-driving' due to specific traffic conditions or rules.
- **Four**: Where vehicles undergo every safety-critical driving functions of different driving situations.
- **Five:** The vehicle is now fully autonomous/self-driving, can operate in any traffic conditions.

In recent times, the automotive industry has been including some aspects of algorithms and mode of operation used for autonomous vehicles into human-driven vehicles, including the four vehicle performance [5]:

- Steering, Acceleration, and De-acceleration of Vehicles.
- Comprehensive Monitoring of the Vehicle Driving environment.
- Reduction in performance of dynamic driving activities.
- Vehicle system capability/capacity (driving modes)

2.1 Effect of Artificial Intelligence on Traffic congestion

Traffic congestion is one of the most well-known road transportation problems in major urban cities all over the world. It is common in every countries of the world, either under-developed, developing or developed countries, it is even more pronounced in developing countries, and developed countries such as United States of America, United Kingdom, and China. China have an all-time high number of traffic congestion in their major cities. Over the years, Transportation engineers have tried to reduce or eradicate traffic congestion in urban areas by constructing more roads and repairing loopholes on highways and introducing road intersections, but these measures have done little or nothing in curbing these problems [6]

Recently, the application of Artificial Intelligence, big data, and autonomous vehicles have proven effective in combating traffic congestion. Artificial Intelligence (AI) tools such as artificial neural network, fuzzy logic model, Genetic algorithm have been successfully applied by researchers in reducing traffic flow of vehicles at freeways, highways and road intersections. It has even gone further in predicting the time of traffic and when a vehicle will arrive at a road intersection. Artificial Intelligence is playing a crucial role in reducing traffic congestion in road transportation systems[6]

The rapid increase of developments in artificial intelligence has been providing unlimited opportunities to improve the efficiency of various types of industries and businesses, mostly in the transportation sector. Artificial Intelligence introduced some innovations such as advanced computational intelligence. Computational intelligence is a process that works or mimic the way the human brain works. There are a lot of challenges facing the transportation sector, but the application of artificial intelligence has gone a long way in reducing these challenges, the most prominent among these challenges are[9]:

- Increasing travel demand.
- CO_2 Emissions.
- Safety concerns.
- Environmental degradation.

Because of the availability of a large amount of quantitative and qualitative big data and also artificial intelligence in this modern age, tackling these challenges more efficiently and effectively has become possible. Various types of artificial intelligence have found their way into the transportation field[9]:

- Artificial Neural Network.
- Genetic Algorithm.
- Simulated Annealing.
- Artificial immune systems.
- Ant Colony Optimization.
- Bee colony optimization.
- Fuzzy logic model

To apply artificial intelligence successfully, excellent understanding is needed on how artificial intelligence and big data are connected and the relationship between transportation systems and transportation variables. The future is looking bright for transportation engineers to determine the method to apply these artificial intelligence technologies to bring about a fast improvement in reducing traffic congestion, decreasing road intersections accidents, making travel time more efficient to their customers and improve the economics and productivity of their important assets[9].

Due to the nature of traffic flow which is stochastic and the short-term prediction that has highly non-linear characteristics, a lot of attention has been received by artificial intelligence and how we can use artificial intelligence as an alternative for traffic flow prediction model. Artificial intelligence is among the best option for modelling and predicting traffic parameters since they can do an approximation of any function with no regards to its degree of non-linearity and also without any preceding knowledge of its functional form [9].

2.2 Signalised Intersection

Intersections are well-known all over the world as part of the most dangerous locations on a highway or a freeway. From transportation reports, almost 43% of accidents that occur on the roads in the USA takes place at or near a signalized road intersections[10] and besides, 40% of road accidents in Norway takes place at road junctions [11]. In Singapore, the annual road accidents statistics indicate that 34.31% of road accidents occur at road intersections [12] and almost 25% of road transportation users lost their lives at road intersections in Canada [13].

There is always a point where the traffic signals on the road will not be able to cope satisfactorily with the increment in the level of traffic intensity on the freeway or highway. Traffic congestion occurs when the time spent by a vehicle that is delayed by traffic lights exceed the time it takes for the traffic lights to control the traffic [14] [15] If this happens the queue length of vehicles may increase, to almost reducing the whole length of the signalized road intersection. Traffic congestion can be divided into 2 categories. They are:

- Systematic (Stable) Traffic congestion.
- Occasional Traffic congestion

Occasional traffic congestion is usually caused by accidents or perhaps unavoidable emergencies. Also, systematic traffic congestion occurs when there is an occasional repetition in time and when there is an occurrence of stability in space. This usually occurs during rush hours. If you want to optimize traffic control, you have to challenge traffic forecasting, recognition, and removal of traffic pre-congestions occurrences; exclusion of roadblocks on the road. The traffic control system of major cities should be able to provide information to the driver if there is traffic congestion on the route the driver wants to pass through and maybe suggest an alternative road for the driver[16]

To increase or improve road safety, improve the efficiency of road transportation networks or to prevent accidents and deaths of people at road intersections, a signalized road intersection must be introduced on the road. Signalized intersections are usually used by developed countries with large urban road transportation network. Usually to reduce traffic congestions and ensure strict adherence to traffic rules by drivers[16].

3.0 Data Collection

The company used as a case study is Mikros Traffic Monitoring (MTM) a subsidiary of Syntell Group Limited, data analyst and traffic engineers at MTM were helpful in the supply of accurate traffic data (Primary and secondary data). The research is focused on traffic flow data on Johannesburg road using loop detectors to determine real time traffic volume and speed level of cars travelling on a defined road, MTM also installed cameras and data loggers at different locations on the highways, freeways and road intersections which take record of the number of vehicles, speed and the capacity of vehicles on various route.

The tools used for the data collection by Mikros Traffic Monitoring (MTM) a subsidiary of Syntell Group Limited, are:

Data loggers: They are electronic equipment used in road transportation networks to record data or information garnered for some time by the movement of vehicles on highways and road intersections. This information gathered can be analyzed or documented for future purposes. They make use of sensors to record information from vehicles.

Loop detectors: They are also called inductive loop detectors. These detectors are installed along with highway bumps or where there is zebra crossing. In addition, they are always quite closed to traffic lights. They detect the number of vehicles passing, the speed and distance covered by the vehicles and the time it took the vehicle to arrive at that certain position on the road.

Video Cameras: These are installed on traffic light poles and usually found around road intersections, road tunnels, highways, freeways and sometimes roundabouts. They are commonly used to catch traffic offenders or monitor the traffic flow of vehicles. The major limitation they have is they are always affected by weather conditions especially during weather or harmattan period, during this period it is was difficult for the video cameras to monitor the traffic flow of vehicles on the road.

Notation

System of case study- Single system = (1)
Available vehicles monitored or screened within the period of investigation=126
Number of light vehicles = 70
Number of large motor vehicles=13
Number of trucks (short)= 20
Number of trucks (medium)=18
Number of trucks (large)=5
System route / road network system =7
Cameras stationed to monitor vehicles = p

Inputs Description for ANN Model
W = Weight or strength of connection
W_{k1} =Weight or strength of connection from neuron 1 to k
W_{k2} =Weight or strength of connection from neuron 2 to k
W_{k3} =Weight or strength of connection from neuron 3 to k
W_{kn}= Connection weight from neuron n to neuron k
X_1 = Input from 1 to neuron k
X_2 = Input from 2 to neuron k
X_n= Input from n to neuron k
j = Index identifying the number of inputs
n = Index identifying destinations
k = Index identifying processing elements
U_k = Linear combiner output due to input signal/Sum of the total neuron

V_k = Linear combiner with bias to support input signal
Y_k = Total output
*Y*pred Predicted value of the neural network model
*Y*exp Experimental value of the neural network model

ANN Model Equation

$$U_K = \sum_{j=1}^{n} x_j w_{kj} \quad (1)$$

$$V_k = U_k + b_k \quad (2)$$

$$V_k = \sum x_j w_{kj} + b_k \quad (3)$$

$$Y_k = S_g F_n (V_k) \quad (4)$$

$$Y_k = \varphi(\cdot) V_k \quad (5)$$

$$Y_k = \varphi(\cdot) \sum_{j=1}^{n} x_j w_{kj} + b_k \quad (6)$$

3.1 Data Extraction

The data was extracted from the data loggers between this dates 15th of July, 2019 to 27th of July, 2019 within 24hours period of time. The vehicles were grouped as light vehicle, light motor vehicle, short truck, medium truck and long truck. About 126 datasets was used which include information like the arrival date, Arrival time, Physical Lane, Logical lane, Vehicle Direction, Class number of Vehicles, Class description of vehicles, Speed of the vehicles and the length (distance) covered by the vehicle. The arrival time of the vehicle is used as the output.

Table 1: A sample of the first five datasets used for the ANN training.

Time	Speed	Length	Physical lane	Logical lane
0.21	93.7	5.23	3	3
0.41	99.6	5.62	6	6
0.45	100.5	4.14	6	6
0.54	121.5	5.43	7	7
0.56	76.3	4.94	3	3

3.2 Results and Discussion

Artificial Neural Network possesses a lot of advantages such as the ability to think for itself than any other artificial intelligence methods and also its accuracy during prediction of time and finding the correlation between input and output of a data analysis. Compared to statistical and analytical software such as Microsoft Excel, AMOS, SPSS and Python, MATLAB (Artificial Neural Network).

The major limitation associated with Artificial Neural Network is the nature of its black box in addition to that there is also the inability of Artificial Neural Network (ANN) framework to understand the cause and effect of independent variable of any input or output. It is quite impossible to know the connecting relationship between variable in an Artificial Neural Network Modelling.
This study is limited by factors such as the composition of vehicles, quality of traffic flow of vehicles during off-peak hours and hourly variation of the traffic flow in 24 hours was not discussed in this study. The traffic data used does not represent the whole of traffic flow on South African Roads, only a section of roads from Johannesburg to Pretoria was used.

3.2.1 Artificial Neural Network (ANN) Model

There are three major types of training algorithm in Artificial Neural Network:

- The Levenberg Marquardt.
- Bayesian Regularization.
- Scaled Conjugate Gradient.

Choosing a suitable training algorithm was dependant on the datasets and the aim to achieve the best possible result. Levenberg-Marquadt was used as the training algorithm because it is fast and easily applicable to the efficient generalization of large datasets Levenberg-Marquadt is the training algorithm that was used. The regression analysis of the training network is shown in Figure 1, the neural network fitting tool was able to train the 126 datasets to achieve a near perfect alignment between the input and output used. This is shown in figure 1 with close relationship between input and output, examined by the results of the regression analysis which is close to 1 (All: R=0.99857)

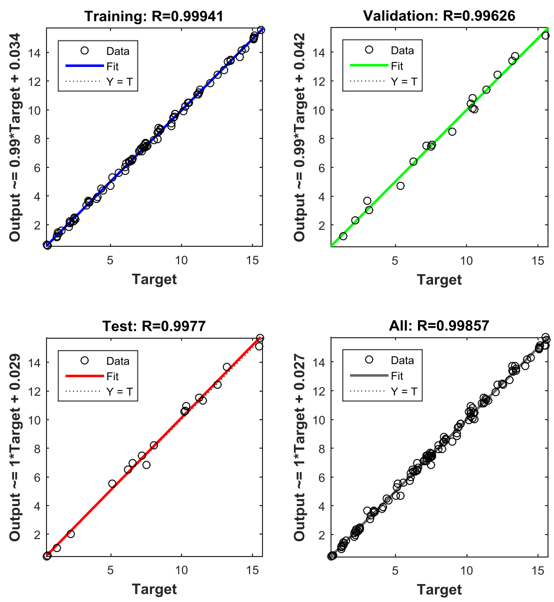

Fig 1: Regression Analysis of the Neural Training Network

A multilayer perceptron (MLP) network was used to predict the Vehicle traffic flow using a three-layer NN model (input, hidden layers and output) with 126 datasets. The datasets comprise of seven attributes, which are the arrival data, arrival time, physical lane, logical lane, vehicle direction, speed of the vehicles and length (distance) covered by the vehicle. The whole datasets were divided into three parts in preparation for training (88), validation (19) and testing (19) respectively. Results showing the training, validation and testing are shown below:

Table 2: Artificial Neural Network Training Results

Results	Target Values	MSE	R

Training	88	2.10946e-2	9.99406e-1
Validation	19	1.17796e-1	9.96261e-1
Testing	19	1.17112e-1	9.97699e-1

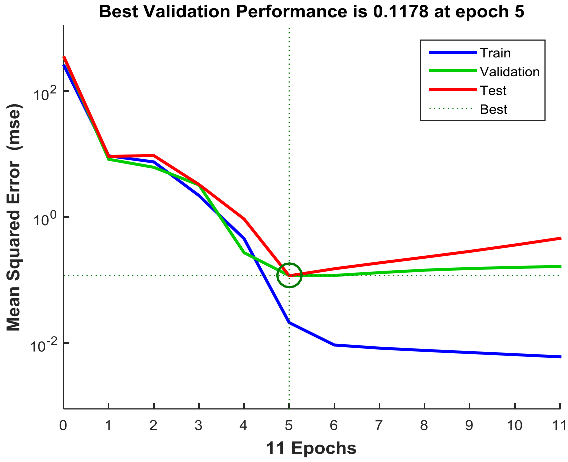

Fig 2: Artificial Network Training Performance

From Figure 2, it can be deduced that the best validation performance is 0.1178 at epoch 5. The point where the validation, testing and best line intersect is at epoch 5 with low mean square error (MSE). The point of intersection is where the best possible artificial neural network result can be obtained.

From Figure 3, the training solutions, testing and validation are in in agreement. The accuracy of the values is a function of low value of MSE as shown in Figure 3.

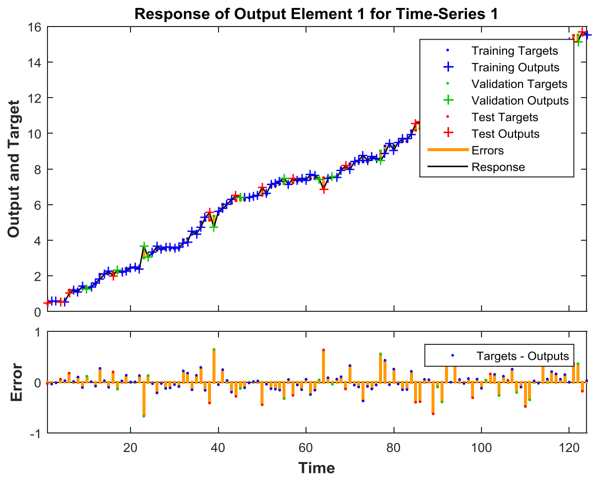

Fig 3: Artificial Network Training Time Series Response

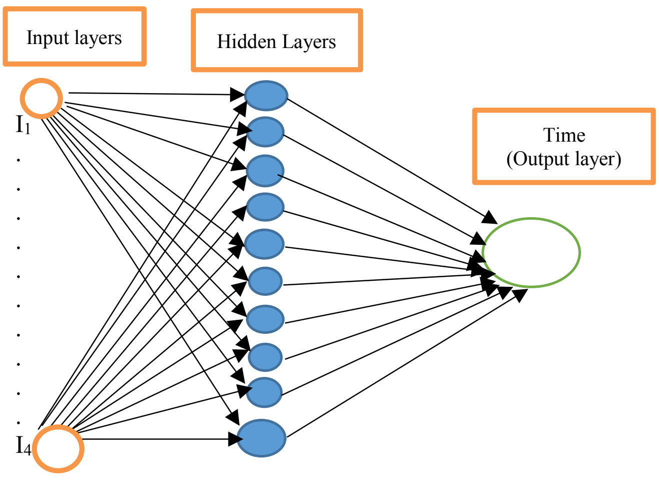

Fig 4: ANN Architecture for best network

For effective prediction of the accuracy of the NN model, training was performed several times by considering one (1) output layer, four (4) input and ten (10) hidden layers. (4-10-1) as shown in the network architecture of Figure 4. During the ANN training, it was deduced that ten (10) hidden neurons produces the best prediction.

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5.0 Conclusion

In this paper, an artificial neural network for prediction of traffic congestion of non-autonomous vehicles at signalized road intersection has been presented. The results clearly show that the ANN model was able to predict the traffic flow of vehicles efficiently considering vehicle speed and class description. To effectively predict the traffic flow of vehicles ten minutes into the future, it is imperative to have a large dataset. ANN will handle complex problem with non-deterministic inputs and large dataset. The limitation of this research is the inability to gain access to robust dataset. With the shift from current revolution to the Fourth Industrial Revolution (4IR), researchers are hopeful to to conduct future research focusing on autonomous vehicles at un-signalized road intersections and application of big data analytics at road intersections.

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